energywave.ai • Research Program and Gates — the protocol, the threads, the commitments
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CONFIDENTIAL — DRAFT — FOR DISCUSSION

Research first, claims last. One protocol, three gates, every thread under it.

The research question, precisely: *do engineered structures — a conditioned solid-state magnetic core, a Casimir cavity, a metal–insulator–metal junction — produce any repeatable, artifact-free electrical output whose energy accounting cannot be closed by known sources?* Expected answer, on the mainstream prior: no. EnergyWave exists to obtain that answer to a standard that survives independent review, and to publish it either way.

THE GATE PROTOCOL

Gate	Question	Pass	Kill
Gate 1 Characterize	Can the reported effects be reproduced, and are they artifacts?	Repeatable output above the fully characterized noise and artifact floor at ≥5σ in blinded runs, surviving every control.	Nothing above the floor, or output explained by a known source. Result published with data.
Gate 2 Conservation test	If something is there, where does the energy come from?	Net output whose closed-system energy balance cannot be closed by known sources, replicated by a second laboratory contracted in advance.	The balance closes, or replication fails. Effect retired publicly.
Gate 3 Extract or retire	Can it be engineered?	Independently validated, repeatable performance under third-party observation — the first point at which EnergyWave would ever state a number. The anti-tamper “paperweight test” is a Gate-3 deliverable.	Scaling, lifetime or stability fails.

Gate names follow the August 2026 white paper (Characterize / Conservation-test / Extract or retire); the September program documents number them Gate 1–3; the October deck shows them as Phases 0–3 with Phase 0 = this round. They are the same three decisions.

ENERGY ACCOUNTING — THE STANDARD EVERY THREAD IS HELD TO

> $E_{\text{all inputs}} = E_{\text{load}} + E_{\text{heat and other outputs}} + \Delta U_{\text{device}}$ — where U includes everything stored in capacitors, batteries, magnetic and strain states and internal thermal reservoirs. DC input is ∫v(t)i(t)dt, never average V × average I. Timing deskew between voltage and current probes in a high-Q circuit can turn a true +50 mW dissipation into a false −75 mW “generation” reading at 1 MHz; calibration must cover the whole measurement chain. A null result bounds the tested configuration at its demonstrated sensitivity — it does not identify an origin. (Physical Architecture & Baseline Model v1.3, §§1–3; Claude review v2, §3.)

THE RESEARCH THREADS — KEPT SEPARATE, HELD TO THE SAME PROTOCOL

Thread	What it is	Evidence to date	Next step
A · Founder's solid-state module Sweet-lineage conditioned ferrite core	Floyd Sweet's vacuum triode amplifier as the named first module (Oct-13 deck). The physical nature of magnet "conditioning" is unestablished; three hypotheses are set out for test: finite stored-energy depletion, open-system environmental transduction, metrology phase skew.	No specimen selected, no hardware ordered, no measured result. Baseline model v1.3 proposes an unconditioned four-step characterization before any conditioning experiment. Specimen, magnetization, winding axis and conditioning procedure are inventor-defined and not yet on record. OPEN	The lead path (founder, 15 Oct 2026). Founder defines the specimen and conditioning **[TBD] — specimen and conditioning**; characterization plan with pass / kill criteria drafted, to be finished with Chip Story.
B · Independent replication MIM diode under a Casimir cavity	Pre-registered replication of Moddel et al. 2021 with cavity / no-cavity twins from the same wafer, sign and magnitude vs. cavity thickness, blinded measurement.	Published claim: 1.4 pW maximum, one lab, >1,000 devices, unreplicated — *Claimed, not replicated*; Maclay 2024 predicts the opposite sign — *Disputed*. Proposed to the Army as an independent evaluation platform (xTech Search 10, submitted 9 Oct 2026).	Qualified independent lead; fabrication access; matched test articles; the artifact-exclusion bench commissioned on known-null samples.
C · The measurement platform Workstream 1B	Shielded isothermal enclosure; calorimetric accounting; lock-in electrical measurement to the picowatt level; blinded sample handling; open data pipeline. Published as a methods paper before any result is claimed.	Design only. Residual value if every thread fails: a credible nanoscale-measurement capability and its publications — stated once, not oversold.	Error budget for external review.

Sweet solid-state is the lead path — the founder's decision of 15 October 2026 (D2, ledger S-44). The September program places the founder's own structure as workstream 1C under the same protocol as the replication; the October deck leads with it; the founder has now confirmed that it leads. A selected lead path is not measured performance and not an operational schedule. Threads B and C remain distinct workstreams, and the xTech entry's proposal of thread B as an independent evaluation capability is not rewritten. Nothing on this page asserts a physical dependency between threads — none has been established — and no thread is described as feeding another. The order on the shared bench and the named measurement lead are **[TBD]**.

EIGHT COMMITMENTS, WRITTEN IN FROM DAY ONE

- Pre-registration of every protocol, public and timestamped.
- Null results published, with data, every time.
- Replication budgeted and contracted before each gate, never assumed.
- External scientific advisory board with veto over any performance statement — at least one declared critic.
- No performance claim before Gate-3 third-party validation.
- Capital released by gate (escrowed tranches on the next round).
- Clean entity separation: EnergyWave, Inc. / McLean Companies LLC / GovThink.
- Open methods: bench design and error-budget methodology published.

WHY PHYSICISTS EXPECT "NO" — SAID HERE BEFORE AN INVESTOR'S PHYSICIST FRIEND SAYS IT

The vacuum is the ground state. Casimir energy is a difference of zero-point energies and the force is conservative, so whatever work it does closing the plates must be paid back to open them. A device delivering steady work from a single equilibrium reservoir is a perpetual-motion machine of the second kind (Brillouin 1950). Dynamical-Casimir photons are real, and the pump pays for them. Even the best-case single-use Casimir stack holds about 2 J/kg against ~9×10^5 J/kg in the lithium-ion cell a humanoid already carries. The program is designed so that if the prior is right, that is learned at Gate 1 for the price of the first tranche — and published.

TECHNICAL DOCUMENTS IN THIS FOLDER

File	Status	Note
EnergyWave_Physical_Architecture_and_Baseline_Model_v1.3_2026-10-05.pdf	ORIGINAL	AI-reviewed working briefing (Antigravity & GPT/Codex; reviewed by Claude). §6 is a historical legal/corporate status table and is superseded by 04_Legal_and_IP.
2026-10-05_energywave-physics-review_claude_v2.md and four companion notes	ORIGINAL	Reservoir inventory; boundary language; Puthoff & Little assessed; Pais / Exodus / regauging checked; GR-coupling evaluation and its review.
EnergyWave_Technical_Prospectus_DRAFT_v1_2026-09-07.docx	HISTORICAL	The September program in full: §4 physics with labels, §6 Gate 1 design and artifact catalogue, §6.5 workstream 1C placeholder, §6.6 decision rule.
Key_Research_Facts_verified_2026-09-07.md	HISTORICAL	One-page corrections to common belief; energy figures used in the narrative.

Sources: Oct-13 deck slides 5–7, 9 (S-01); Technical Prospectus v1 §§4, 6–7 (M-05); Program_Summary_and_Gates.md and Key_Research_Facts (M-03b); DoD white paper v2.0 §3 (M-07); Physical Architecture & Baseline Model v1.3 (S-11/M-14); Claude physics review v2 and source notes (S-13, S-15–S-18); xTech white paper (M-10); founder decisions of 15 Oct 2026 (S-44).